Tax Evasion and Productivity

Hans Martinez

Western University

October 21, 2026

Tax evasion through input overreporting

- VAT allows firms to deduct taxes paid on their purchases from the taxes collected on sales
- By overreporting their inputs, firms can reduce their tax bill
- Input overreporting is widespread: a common practice and a concern among OECD countries (OECD 2017)
  - Losses of 1.2–2.7% of total tax revenue in Chile, Colombia, Ecuador, Poland and Slovakia
  - Conditional on detected evasion: true losses are plausibly larger
- Governments have limited resources, and overreporting costs the government tax revenue

Question: Can we detect and measure tax evasion through input overreporting? How much does overreporting of inputs cost the government?

Why is it difficult to detect?

- Overreporting is unobserved: firms conceal their actions because it is illegal
- Tax-evading firms might *pass* as low-productivity firms
- For a given level of output, high input use by a firm might be because the firm is overreporting or because its productivity is low

What have people done?

- **Transaction-level data and administrative records** (Carrillo et al. 2022; Zumaya et al. 2021)
  - False invoicing, ghost-firm invoices, invoice mills
  - Access to this data is restricted; it is conditional on detected evasion
- **On the consumer side:** a benchmark group to measure misreporting from potentially evading individuals (Pissarides and Weber 1989; Gorodnichenko et al. 2009)
  - Self-employed vs. employees
  - The self-employed might underreport their income to pay less income tax; employees cannot misreport because their income is reported by their employers
  - Consumption is a function of true income: comparing their consumption backs out how much the self-employed underreport

The natural analogue for firms: output

- Output is a function of true inputs (assuming output is correctly reported)
- If we knew the production function (PF) and productivity, we could back out true inputs: the difference between reported and true inputs is the overreporting
- However, we rely on *reported inputs* to recover the PF and productivity
  - Proxy-variable methods rely on the *flexible* input (Levinsohn and Petrin 2003; Ackerberg et al. 2015; Gandhi et al. 2020)
    - If the flexible input is deductible, firms might overreport it: materials, in my application
- We have to solve the problem simultaneously, for overreporting and productivity

What do I do?

- Instead of looking into the PF equation, I take the FOC of the firm's profit-maximization problem
- The FOC links the materials share of revenue to the materials elasticity of output, up to measurement error
  - It does not depend on productivity (Hicks-neutral productivity)
- Just like in the individuals' case, I estimate this elasticity from a benchmark group of firms that do not overreport
  - The benchmark group can have different productivity than the potentially tax-evading firms
  - The critical assumption: both share a common technology
- With the elasticity at hand, the difference between reported and true materials reveals the overreporting, up to measurement error

What do I find?

Colombian manufacturing, 1981–1991:

1. **Detect:** overreporting in 9 of the 18 industries liable for the sales tax (about two fifths of manufacturing output); none in the 2 exempt food industries
2. **Measure:** potentially tax-evading firms overreport on average 4.5–23.1% of their true materials, depending on the industry
3. **Cost:** undetected overreporting costs 4.7% of gross sales-tax revenue; a 1% increase in the rate on purchases lowers net sales-tax revenue by about 1.7% of gross sales-tax revenue
4. **Productivity:** failing to correct for overreporting leads to biased estimates of the PF and productivity

Setting: Colombia, 1981–1991

- Annual Survey of Manufacturing (EAM): a panel of manufacturing plants with more than 10 employees
  - Output and inputs, juridical organization, and the sales tax charged on sales and paid on purchases
- The Colombian sales tax worked as a VAT: manufacturers credited the tax paid on their purchases against the tax charged on their sales
- The credit rewards overreporting materials: each fictitious peso lowers the tax liability by the rate on purchases
  - Capital goods could not be credited and labour was not a taxed purchase
- Weak enforcement: tax evasion flourished, and the tax authority audited inefficiently (Sánchez Torres and Gutiérrez Sourdis 1994; Perry and Orozco de Triana 1990)
- Food products (311, 312) were exempt from the sales tax: almost no reward to overreporting

The benchmark group: corporations

- Two kinds of scrutiny restrained corporations from overreporting: government and market
  - Government: the Superintendent of Corporations closely supervised corporations and required each to have an auditor (Joint Tax Program of the Organization of American States and the Inter-American Development Bank 1965)
  - Market: tradable shares subject corporations to market discipline; the market penalizes aggressive tax behaviour even when it is legal (Hanlon and Slemrod 2009)
- Limited liability companies (LLCs) and proprietorships faced only the tax authority, whose audits were weak
- Ecuador: tax evaded through fictitious purchases was 1.7% for corporations and 11.5% for sole proprietorships (Carrillo et al. 2022)
  - If corporations nonetheless overreport, my estimates are a lower bound
- Corporations have different productivities than the rest of the firms. But that's OK: the critical assumption is a common technology

Model: firms, taxes and overreporting

- Price-taking firms maximize expected after-tax profits; they share a common technology but differ in productivity

$$Y_{it} = K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it} + \varepsilon_{it})$$

  - ω_{it}: productivity, known to the firm; ε_{it}: measurement error in output
- Materials are flexible and deductible; firms report $M_{it}^{*} = M_{it} + e_{it}$, $e_{it} \geq 0$
  - Overreporting is detected with a probability $q(e)$ and costs $\kappa(e, \omega)$; both are zero for a firm that does not overreport

Model: the firm's problem

$$\max_{M,e,K,L} \mathbb{E}[\pi_{it}] = (1-\tau_S)P_t E_\varepsilon[K_{it}^{\alpha_K} L_{it}^{\alpha_L} M_{it}^{\beta} \exp(\omega_{it}+\varepsilon_{it})] - (1-\tau_P)\rho_t M_{it} + \tau_P \rho_t e_{it}[1-(1+\phi)q(e_{it})] - \kappa_{it}(e_{it},\omega_{it})$$

- Revenue net of the tax on sales; the cost of true materials net of the credit on them; the rental cost of capital and the wage bill
- The credit on overreported materials net of the expected repayment and fine; the cost of evasion
- Underreporting would increase the tax liability: no firm has an incentive to do so, $e_{it} \geq 0$

Model: the materials FOC

- The FOC with respect to materials gives the log share of materials in revenue, net of sales taxes:

$$\ln\left(\frac{(1-\tau_P)\rho_t M_{it}}{(1-\tau_S)P_t Y_{it}}\right) = \ln\beta - \varepsilon_{it}$$

  - Productivity does not appear in the share (Hicks-neutral productivity)
- With reported materials M^*_{it}, the log share s_{it} satisfies

$$s_{it} = \ln\beta - \varepsilon_{it} + u_{it}, \qquad u_{it} = \ln(M^*_{it}/M_{it}) \geq 0$$

Testing for overreporting

- Corporations do not overreport: $s_{it} = \ln \beta - \varepsilon_{it}$ gives $\ln \beta = E[s_{it} \mid \text{corporations}]$
- Unincorporated firms: the observed residual $\mathcal{V}_{it} = s_{it} - \ln \beta = u_{it} - \varepsilon_{it}$

$$H_0 : \mu_{\mathcal{V}} = 0 \qquad \text{vs.} \qquad H_1 : \mu_{\mathcal{V}} > 0$$

- One-sided: no firm has an incentive to underreport deductible inputs. A rejection can only point to overreporting
- The test does not depend on the functional forms of the detection probability and the cost of evasion, nor on the Cobb-Douglas form of the technology
- Confidence set for $\mu_{\mathcal{V}}$ by test inversion, clustered by plant

Testing results

Industry	$\hat{\mu}$	95% region	Industry	$\hat{\mu}$	95% region
311 Food products	0.024	$(\cdot, 0.048]$	**342 Printing and publishing**	0.074	[0.036, 0.112]
312 Other food products	−0.018	$(\cdot, 0.076]$	**351 Industrial chemicals**	0.165	[0.001, 0.329]
313 Beverages	0.390	[0.310, 0.470]	**352 Other chemicals**	0.057	[0.002, 0.111]
321 Textiles	0.151	[0.115, 0.188]	356 Plastic products	−0.107	†
322 Wearing apparel	0.224	[0.195, 0.253]	**369 Non-metallic minerals**	0.193	[0.109, 0.277]
323 Leather products	−0.192	†	381 Metal products	−0.005	$(\cdot, 0.025]$
324 Footwear	0.095	[0.059, 0.131]	382 Non-electrical machinery	−0.014	$(\cdot, 0.026]$
331 Wood products	0.250	[0.175, 0.325]	383 Electrical machinery	0.034	$(\cdot, 0.089]$
332 Furniture	−0.007	$(\cdot, 0.059]$	384 Transport equipment	−0.129	†
341 Paper products	−0.066	$(\cdot, 0.004]$	390 Other manufacturing	0.057	$(\cdot, 0.115]$

$\hat{\mu}$: mean of $\hat{\nu}_{it}$ among unincorporated firms. 95% test-inversion region (sharp test); bold: the region excludes 0 (overreporting detected). $(\cdot, b]$: 0 not rejected. †: empty region.

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- The test detects overreporting in 9 of the 18 industries liable for the sales tax: 4 of the 5 largest by output; 41% of manufacturing output
- No overreporting in either exempt industry, Food products (311) and Other food products (312): this is expected

Counterfactual: the rate on purchases

- How does net sales-tax revenue change as the government changes the sales-tax rate on purchases?
- A change in the rate moves net sales-tax revenue through three channels:
  1. Mechanical: every peso the firm claims is credited at the new rate
  2. Overreporting: a higher rate raises the marginal benefit of overreporting
  3. Materials: the rate changes the net-of-tax price of materials relative to output
- The overreporting response needs the model: the FOC with respect to e, the evasion condition

$$\underbrace{\tau_P \rho_t \left[1 - (1+\phi)\left(q(e_{it}) + q'(e_{it}) e_{it}\right)\right]}_{\text{marginal benefit}} = \underbrace{\frac{\partial \kappa_{it}(e_{it}, \omega_{it})}{\partial e_{it}}}_{\text{marginal cost}}$$

  - A higher tax rate on purchases raises overreporting
  - The cost of evasion depends on productivity, which lets the model match the observed pattern in recent evidence (Carrillo et al. 2022)

One unobservable: true materials

- I cast the estimation problem in terms of a single unobservable, true materials M:

$$e(M) = M^*_{it} - M, \qquad \varepsilon(M) = \ln(M^*_{it}/M) - \mathcal{V}_{it},$$

$$\omega(M) = \tilde{\mathcal{W}}_{it} - (1-\beta)\varepsilon(M)$$

  - $\mathcal{V}_{it}$: the observed residual of the test; $\tilde{\mathcal{W}}_{it}$: from the production-function estimates (appendix)
  - The evasion condition then gives the cost shock $\psi(M)$
- Moments from the model's independence assumptions, e.g. $E[\psi] = 0$, $E[\psi \ln \tau_P] = 0$, $E[\varepsilon] = 0$
- Why ELVIS?
  - Moments are nonlinear in the unobservable, true materials. So a standard GMM or instruments/2SLS would not work
  - We could assume a parametric distribution (indirect inference, maximum simulated likelihood). ELVIS does not impose parametric assumptions on the distribution of the unobservables

ELVIS (Schennach 2014)

- Entropic Latent Variable Integration via Simulation
- A method of simulated moments: MCMC methods sample from a non-parametric distribution of the unobservables conditional on the observables, such that the moment conditions hold
- It turns an existence question into an optimization problem:

$$\inf_{f_{M|Z}} \left\| E[g(M,Z,\theta)] \right\| = 0 \iff \inf_{\gamma} \left\| E[\tilde{g}(Z,\theta,\gamma)] \right\| = 0, \qquad \tilde{g}(Z,\theta,\gamma) = \frac{\int g\, e^{\gamma' g} f^0_M}{\int e^{\gamma' g} f^0_{M|}}$$

  - "Is there a distribution of the unobservables conditional on the observables such that the moments hold?" becomes "what is the distribution of the unobservables conditional on the observables that maximizes entropy?"
- Why do we care about entropy? We don't. We just need a distribution that satisfies the moment conditions; ELVIS selects the one that maximizes entropy
- ELVIS is always going to find a distribution: we test whether the moments are equal to zero, $TS_n = 2n\hat{L}_n(\theta)$ against $\chi^2_{d_g}$

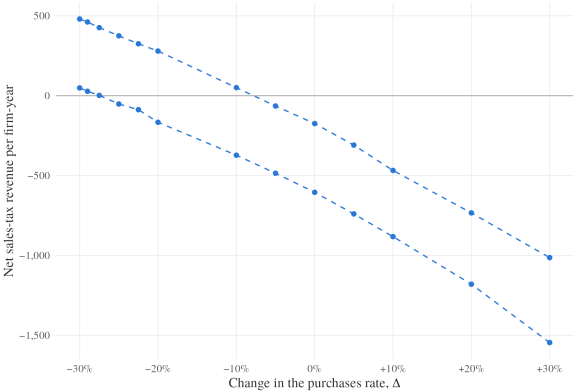
The counterfactual as a moment

- If we can write our counterfactual as a moment condition, then we can use ELVIS to answer the counterfactual
- Net sales-tax revenue at the new rate $\tilde{\tau}_P = (1 + \Delta)\tau_P$ depends on the latent M_{it}:

$$R_i(\Delta; M_i) = \tau_S P_t Y_{it}(\Delta) - \tilde{\tau}_P \big[M_{it}(\Delta) + (1 - q(\tilde{e}_i))\tilde{e}_i\big]$$

- Add the counterfactual moment, with an auxiliary parameter T, to the set of moments: $E\big[R_i(\Delta; M_i) - T\big] = 0$
- For each T: is there a distribution of the unobservables conditional on the observables that satisfies the model's moments AND the counterfactual moment? Yes or no
  - The confidence set collects the values of T that the test does not reject
- Why a set: the entropy-maximizing distribution should not be treated as the actual distribution of M, since many conditional distributions of M given Z may satisfy the moment condition (Schennach 2022)
  - "Latent variable models are often set-identified" (Schennach 2014, 346)

Results: net sales-tax revenue



Real pesos per firm-year, unincorporated firms in the 9 industries where the test rejects

- Current rate: -391 $[-605, -174]$: refunds on claimed deductions exceed gross sales-tax revenue
- Breaks even at a cut of about 17%; significantly positive from a cut of 27.5% $[2, 426]$
- A 1% increase lowers net sales-tax revenue by about 1.7% of gross sales-tax revenue $[1.47, 1.91]$

Results: the cost of overreporting

- Undetected overreporting costs 4.7% $[2.8, 7.1]$ of gross sales-tax revenue from the unincorporated firms in the 9 industries
- At least 1.5% $[1.1, 3.1]$ across all firms in the sample
- Same order of magnitude as what tax authorities have detected from input overreporting: 1.2–2.7% of total tax revenue in five countries (OECD 2017; Carrillo et al. 2022; CASE and IHS 2018; Longinotti 2022)

Conclusion

- Can we detect and measure tax evasion through input overreporting? How much does it cost the government?
- Overreporting can be detected and measured from production data and a benchmark group alone
- It is common and large
- It is costly in two ways: for the government, which credits fictitious purchases, and for productivity measurement
- A higher rate on purchases raises claimed deductions by more than its mechanical effect: net sales-tax revenue falls by more than the higher credit alone implies

Appendix

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